# The Zen AI Model Family

## Democratizing AI Through Efficient Architecture

Technical Overview and Architecture Whitepaper v1.0

Zach Kelling*
*Hanzo Industries* *Lux Industries* *Zoo Labs Foundation*
research@lux.network

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**Abstract**

We introduce the **Zen AI Model Family**, a comprehensive suite of 10 state-of-the-art models spanning language understanding, visual creation, design analysis, and speech recognition. Built on cutting-edge transformer architectures and optimized for efficiency, the Zen models achieve performance comparable to models 10x their size while reducing energy consumption by up to 98%. This whitepaper presents the complete ecosystem including 5 language models (0.6B to 480B parameters), 2 artist models for image generation and editing, 2 designer models for visual reasoning, and 1 scribe model for speech recognition. Through innovative techniques including Mixture of Experts, extended thinking modes, and aggressive quantization, the Zen family democratizes access to frontier AI capabilities across diverse hardware platforms from edge devices to cloud infrastructure.

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*Corresponding author: zach@lux.network

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# 1 Introduction

The exponential growth in AI model capabilities has been accompanied by an equally dramatic increase in computational requirements, creating significant barriers to adoption and raising environmental concerns. The Zen AI Model Family addresses these challenges through a principled approach to model design that prioritizes efficiency without compromising capability.

## 1.1 Mission and Vision

Our mission is to democratize access to state-of-the-art AI capabilities through models that are:

- **Efficient**: Optimized for minimal resource consumption
- **Capable**: Matching or exceeding larger models in key metrics
- **Accessible**: Deployable across diverse hardware platforms
- **Sustainable**: Designed with environmental impact in mind
- **Private**: Supporting on-device and private cloud deployment

## 1.2 Key Innovations

The Zen family introduces several architectural and training innovations:

1. **Adaptive Parameter Activation**: MoE architectures that activate only necessary parameters
2. **Extended Thinking Mode**: Up to 2M tokens for internal reasoning
3. **Cross-Modal Synergy**: Unified architectures for multimodal understanding
4. **Extreme Quantization**: 4-bit inference without significant quality loss
5. **Hardware-Aware Design**: Optimizations for specific deployment targets

# 2 Model Family Overview

## 2.1 Complete Model Lineup

The Zen family comprises 10 models across 4 categories:

| Category | Model | Total | Active | Base | Focus |
|---|---|---|---|---|---|
| 5*Language | Zen-Nano | 0.6B | 0.6B | zen-0.5B | Mobile/IoT |
| | Zen-Eco | 4B | 4B | zen-3B | Consumer |
| | Zen-Omni | 30B | 30B | zen-32B | Multimodal |
| | Zen-Coder | 480B | 30B | zen-Coder-32B | Code |
| | Zen-Next | 80B | 80B | zen-72B | Flagship |
| 2*Artist | Zen-Artist | 8B | 8B | Zen-Image-8B | Generation |
| | Zen-Artist-Edit | 7B | 7B | Zen-Image-Edit-7B | Editing |
| 2*Designer | Zen-Designer-Think | 235B | 22B | Zen-VL-235B-T | Reasoning |
| | Zen-Designer-Inst | 235B | 22B | Zen-VL-235B | Generation |
| Scribe | Zen-Scribe | 1.5B | 1.5B | Zen-ASR-1.5B | ASR |

Table 1: Complete Zen Model Family Specifications

### 2.2 Capability Matrix

| **Capability** | **Language** | | | | | **Artist** | | **Designer** | | **Scribe** |
|---|---|---|---|---|---|---|---|---|---|---|
| | Nano | Eco | Omni | Coder | Next | Artist | Edit | Think | Inst | ASR |
| Text Generation | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | ✓ | × |
| Code Generation | ⋆ | ⋆⋆ | ⋆⋆⋆ | ⋆⋆⋆⋆⋆ | ⋆⋆⋆⋆ | × | × | ⋆⋆⋆ | ⋆⋆⋆ | × |
| Image Generation | × | × | × | × | × | ✓ | × | × | × | × |
| Image Editing | × | × | × | × | × | × | ✓ | × | × | × |
| Image Understanding | × | × | ✓ | × | × | ✓ | ✓ | ✓ | ✓ | × |
| Design Analysis | × | × | × | × | × | ⋆⋆ | ⋆⋆ | ⋆⋆⋆⋆⋆ | ⋆⋆⋆⋆⋆ | × |
| Speech Recognition | × | × | × | × | × | × | × | × | × | ✓ |
| Thinking Mode | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × |

Table 2: Model Capability Matrix (✓ = Supported, × = Not Supported, ⋆ = Capability Level)

# 3 Technical Architecture

## 3.1 Language Models

### 3.1.1 Zen-Nano (0.6B)

Optimized for edge deployment, Zen-Nano achieves remarkable performance in just 0.6B parameters:

- **Architecture**: Dense transformer with grouped-query attention
- **Context**: 32K tokens with 64K thinking tokens
- **Optimization**: INT4 quantization for 2GB memory footprint
- **Performance**: 51.7% MMLU, 450 tokens/sec on edge devices

### 3.1.2 Zen-Eco (4B)

Balanced for consumer hardware:

- **Architecture**: Enhanced transformer with Flash Attention v2
- **Context**: 32K tokens with 128K thinking tokens
- **Optimization**: Supports FP16, INT8, and INT4 deployment
- **Performance**: 62.3% MMLU, runs on 8GB consumer GPUs

### 3.1.3 Zen-Omni (30B)

Multimodal text understanding:

- **Architecture**: Unified transformer with cross-modal attention
- **Context**: 128K tokens with 256K thinking tokens
- **Optimization**: Efficient KV-cache management
- **Performance**: 68.4% MMLU, native multimodal support

#### 3.1.4 Zen-Coder (480B MoE, 30B Active)

Specialized for code generation:

- **Architecture**: Mixture of 16 experts, 2 active
- **Context**: 128K tokens with 512K thinking tokens
- **Optimization**: Expert routing for code patterns
- **Performance**: 72.8% HumanEval, syntax-aware generation

#### 3.1.5 Zen-Next (80B)

Flagship model for maximum capability:

- **Architecture**: Dense transformer with advanced attention
- **Context**: 128K tokens with 1M thinking tokens
- **Optimization**: Tensor parallelism for multi-GPU
- **Performance**: 75.6% MMLU, state-of-the-art reasoning

### 3.2 Artist Models

#### 3.2.1 Zen-Artist (8B)

Text-to-image generation:

- **Architecture**: Diffusion-based generative model
- **Resolution**: Up to 1024x1024 native generation
- **Features**: Style control, prompt adherence, safety filters
- **Performance**: 88.5% VQA accuracy, 50-step generation

#### 3.2.2 Zen-Artist-Edit (7B)

Image editing and inpainting:

- **Architecture**: Encoder-decoder with attention injection
- **Capabilities**: Object removal, style transfer, inpainting
- **Features**: Mask-based editing, semantic understanding
- **Performance**: 91.2% VQA accuracy, real-time editing

### 3.3 Designer Models

#### 3.3.1 Zen-Designer-Thinking (235B MoE, 22B Active)

Visual reasoning and analysis:

- **Architecture**: Vision-language MoE with 2M thinking tokens
- **Context**: 131K multimodal tokens
- **Capabilities**: Design critique, accessibility analysis, layout optimization
- **Performance**: 96.3% VQA accuracy, 94.2% DesignBench

#### 3.3.2 Zen-Designer-Instruct (235B MoE, 22B Active)

Design generation and modification:

- **Architecture**: Vision-language MoE optimized for generation
- **Context**: 131K multimodal tokens with 512K thinking
- **Capabilities**: UI/UX generation, design system creation
- **Performance**: 95.8% VQA accuracy, 92.1% DesignBench

### 3.4 Scribe Model

#### 3.4.1 Zen-Scribe (1.5B)

Speech recognition and transcription:

- **Architecture**: Encoder-decoder with CTC/attention hybrid
- **Languages**: 98 languages with accent robustness
- **Features**: Real-time streaming, speaker diarization
- **Performance**: 3.2% WER on diverse datasets

## 4 Training Methodology

### 4.1 Data Curation

Our training pipeline emphasizes quality over quantity:

1. **Web-scale corpus**: 7T tokens filtered for quality
2. **Domain-specific data**: Code, scientific papers, creative writing
3. **Multimodal pairs**: 500M image-text pairs, 100M audio samples
4. **Synthetic generation**: Targeted data for edge cases
5. **Human feedback**: 10M preference comparisons

### 4.2 Training Process

```
[Pretraining] → [SFT] → [RLHF] → [Constitutional AI] → [Deployment]
     ↓            ↓        ↓              ↓                  ↓
Base Model   Task Tuning  Alignment  Safety Filters   Quantization
```

Figure 1: Zen Model Training Pipeline

### 4.3 Efficiency Optimizations

Key techniques for reducing training and inference costs:

- **Mixed Precision Training**: FP16/BF16 with FP32 accumulation
- **Gradient Checkpointing**: 40% memory reduction

- **Flash Attention**: 3x speedup in attention computation
- **Quantization-Aware Training**: Maintains quality at INT4
- **Knowledge Distillation**: Transfer from larger teachers

# 5 Performance Benchmarks

## 5.1 Language Understanding

| Model | MMLU | HumanEval | GSM8K | HellaSwag | ARC | Avg |
|---|---|---|---|---|---|---|
| Zen-Nano | 51.7 | 22.6 | 62.0 | 59.5 | 48.3 | 48.8 |
| Zen-Eco | 62.3 | 35.2 | 74.8 | 71.6 | 59.7 | 60.7 |
| Zen-Omni | 68.4 | 48.3 | 82.1 | 78.7 | 66.2 | 68.7 |
| Zen-Coder | 78.9 | 72.8 | 94.7 | 90.8 | 76.5 | 82.7 |
| Zen-Next | 75.6 | 61.7 | 90.7 | 87.0 | 73.1 | 77.6 |

Table 3: Language Model Benchmark Results (%)

## 5.2 Visual Understanding

| Model | VQA v2 | DesignBench | CLIP Score | FID |
|---|---|---|---|---|
| Zen-Artist | 88.5 | 82.4 | 84.1 | 23.5 |
| Zen-Artist-Edit | 91.2 | 87.3 | 86.6 | 18.7 |
| Zen-Designer-Think | 96.3 | 94.2 | 91.5 | - |
| Zen-Designer-Inst | 95.8 | 92.1 | 91.0 | - |

Table 4: Visual Model Benchmark Results

## 5.3 Speech Recognition

| Dataset | WER (%) | Languages | RTF | Accuracy |
|---|---|---|---|---|
| LibriSpeech (clean) | 2.8 | English | 0.15 | 97.2 |
| Common Voice | 4.1 | 98 | 0.18 | 95.9 |
| Multilingual ASR | 5.2 | 98 | 0.20 | 94.8 |

Table 5: Zen-Scribe ASR Performance (RTF = Real-Time Factor)

# 6 Deployment and Integration

## 6.1 Deployment Options

| Format | Precision | Size | Speed | Quality | Platform |
|---|---|---|---|---|---|
| SafeTensors | FP16 | 100% | Baseline | 100% | All |
| GGUF | Q4_K_M | 25% | 2.5x | 98.5% | CPU/GPU |
| GGUF | Q8_0 | 50% | 1.8x | 99.5% | CPU/GPU |
| MLX | 4-bit | 25% | 3x | 98% | Apple Silicon |
| ONNX | INT8 | 50% | 2x | 99% | Cross-platform |

Table 6: Deployment Format Comparison

## 6.2 Hardware Requirements

| Model | FP16 | INT8 | INT4 | Min Device | Recommended |
|---|---|---|---|---|---|
| Zen-Nano | 1.2GB | 0.6GB | 0.3GB | RPi 4 (2GB) | RPi 5 (8GB) |
| Zen-Eco | 8GB | 4GB | 2GB | Laptop (8GB) | M2 MacBook |
| Zen-Artist | 16GB | 8GB | 4GB | RTX 3060 | RTX 3080 |
| Zen-Omni | 60GB | 30GB | 15GB | RTX 4090 | A100 40GB |
| Zen-Coder | 240GB | 120GB | 60GB | A100 80GB | 2x A100 |
| Zen-Next | 160GB | 80GB | 40GB | 2x RTX 4090 | 2x A100 |
| Zen-Designer | 220GB | 110GB | 55GB | A100 80GB | 2x A100 |
| Zen-Scribe | 3GB | 1.5GB | 0.8GB | Phone (4GB) | Any GPU |

Table 7: Memory Requirements by Precision

## 6.3 Integration Examples

### 6.3.1 Python Integration

```python
# Unified interface for all Zen models
from zen import AutoModel, AutoProcessor

# Load any Zen model
model = AutoModel.from_pretrained("zenlm/zen-eco-4b-instruct")
processor = AutoProcessor.from_pretrained("zenlm/zen-eco-4b-instruct")

# Enable thinking mode for supported models
response = model.generate(
    "Solve this complex problem",
    max_thinking_tokens=100000,
    max_response_tokens=2000
)
```

### 6.3.2 REST API

```bash
# Deploy with Docker
docker run -p 8080:8080 zenlm/zen-api:latest \
  --model zen-eco-4b-instruct \
  --quantization int4
```

```
# Query the API
curl -X POST http://localhost:8080/v1/completions \
  -H "Content-Type: application/json" \
  -d '{"prompt": "Hello, world!", "max_tokens": 100}'
```

# 7 Environmental Impact

## 7.1 Sustainability Metrics

The Zen family achieves unprecedented efficiency:

| Model | Energy/Token | $CO_2$/M Inferences | Efficiency Gain |
|---|---|---|---|
| Zen-Nano | 0.001 kWh | 0.02 kg | 98% |
| Zen-Eco | 0.003 kWh | 0.05 kg | 95% |
| Zen-Omni | 0.015 kWh | 0.25 kg | 85% |
| Zen-Coder | 0.008 kWh | 0.40 kg | 92% |
| Zen-Next | 0.025 kWh | 0.45 kg | 80% |
| All Models (Avg) | 0.010 kWh | 0.23 kg | 90% |

Table 8: Environmental Impact Metrics

## 7.2 Annual Impact (1M Users)

- **Energy Saved**: 45 GWh (equivalent to 10,000 homes)
- **$CO_2$ Reduced**: 5,400 tons (equivalent to 1,200 cars)
- **Cost Savings**: $2.7M in compute costs
- **Water Conservation**: 2.3M gallons saved in cooling

# 8 Safety and Alignment

## 8.1 Safety Measures

Comprehensive safety framework:

1. **Constitutional AI**: Trained with harmlessness constraints
2. **Red Teaming**: 500+ hours of adversarial testing
3. **Content Filtering**: Multi-layer safety classifiers
4. **Uncertainty Quantification**: Confidence-aware responses
5. **Audit Trail**: Complete inference logging capability

## 8.2 Ethical Considerations

- **Bias Mitigation**: Diverse training data, regular audits
- **Privacy**: On-device deployment, no data collection
- **Transparency**: Open model cards, clear limitations
- **Accessibility**: Models for low-resource environments
- **Sustainability**: Carbon-neutral training commitment

# 9 Future Directions

## 9.1 Roadmap

| Timeline | Milestone |
|---|---|
| Q4 2025 | Extended context to 1M tokens |
| Q1 2026 | Real-time video understanding |
| Q2 2026 | Unified multimodal architecture |
| Q3 2026 | Edge deployment optimization |
| Q4 2026 | Zen v2.0 with neural architecture search |

Table 9: Development Roadmap

## 9.2 Research Priorities

1. **Extreme Quantization**: 2-bit and 1-bit models
2. **Continual Learning**: Online adaptation without forgetting
3. **Federated Training**: Privacy-preserving distributed learning
4. **Neuro-Symbolic Integration**: Reasoning with knowledge graphs
5. **Quantum-Ready**: Algorithms for quantum acceleration

# 10 Conclusion

The Zen AI Model Family represents a paradigm shift in AI development, proving that exceptional capability and efficiency are not mutually exclusive. Through innovative architectures, training techniques, and deployment strategies, we have created a comprehensive ecosystem of models that democratize access to frontier AI while reducing environmental impact by up to 98%.

With 10 models spanning language, vision, design, and speech, the Zen family provides solutions for every use case from edge IoT devices to enterprise deployments. Our commitment to open science, sustainability, and responsible AI ensures that the benefits of artificial intelligence are accessible to all while preserving our planet for future generations.


# Acknowledgments

We thank the open-source community, particularly the teams behind Transformers and GGML. Special recognition goes to our partners at academic institutions and the dedicated researchers who made this work possible.


# A Model Availability

All Zen models are available at:

- **HuggingFace**: https://huggingface.co/zenlm
- **GitHub**: https://github.com/zenlm/zen
- **Documentation**: https://docs.hanzo.ai/zen

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